```
!pip install transformers==4.41.2
```

```
Collecting transformers==4.41.2
  Downloading transformers-4.41.2-py3-none-any.whl.metadata (43 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 43.8/43.8 kB 1.9 MB/s eta 0:00:00
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (3.29.0)
Collecting huggingface-hub<1.0,>=0.23.0 (from transformers==4.41.2)
  Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (26.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (202
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (2.32.4)
Collecting tokenizers<0.20,>=0.19 (from transformers==4.41.2)
  Downloading tokenizers-0.19.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.7 kB)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (0.
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.41.2) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.23.
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transform
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.41.2)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4
Downloading transformers-4.41.2-py3-none-any.whl (9.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 9.1/9.1 MB 64.7 MB/s eta 0:00:00
Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 566.4/566.4 kB 33.5 MB/s eta 0:00:00
Downloading tokenizers-0.19.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.6/3.6 MB 72.2 MB/s eta 0:00:00
Installing collected packages: huggingface-hub, tokenizers, transformers
  Attempting uninstall: huggingface-hub
    Found existing installation: huggingface_hub 1.11.0
    Uninstalling huggingface_hub-1.11.0:
      Successfully uninstalled huggingface_hub-1.11.0
  Attempting uninstall: tokenizers
    Found existing installation: tokenizers 0.22.2
    Uninstalling tokenizers-0.22.2:
      Successfully uninstalled tokenizers-0.22.2
  Attempting uninstall: transformers
    Found existing installation: transformers 5.8.0
    Uninstalling transformers-5.8.0:
      Successfully uninstalled transformers-5.8.0
Successfully installed huggingface-hub-0.36.2 tokenizers-0.19.1 transformers-4.41.2
WARNING: The following packages were previously imported in this runtime:
  [tokenizers,transformers]
You must restart the runtime in order to use newly installed versions.
```

RESTART SESSION

```
import tensorflow as tf
from transformers import BertTokenizer, TFAutoModelForSequenceClassification
```

```
The cache for model files in Transformers v4.22.0 has been updated. Migrating your old cache. This is a one-time only operat
```

0/0 [00:00<?, ?it/s]

```
texts = [
    "I love this product!",
    "This is the worst experience ever."
]

labels = [1, 0]  # 1 = Positive, 0 = Negative
```

```
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: FutureWarning: `resume_download` is deprecated
  warnings.warn(
```

config.json: 100% 570/570 [00:00<00:00, 37.0kB/s]

```
encodings = tokenizer(
    texts,
    padding=True,
    truncation=True,
    max_length=32,
    return_tensors="tf"
)
```

```
labels = tf.convert_to_tensor(labels)
```

```
model = TFAutoModelForSequenceClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=2
)
```

```
model.safetensors: 100%                                    440M/440M [00:04<00:00, 108MB/s]
All PyTorch model weights were used when initializing TFBertForSequenceClassification.

Some weights or buffers of the TF 2.0 model TFBertForSequenceClassification were not initialized from the PyTorch model and
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

```
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=5e-5),
    loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=["accuracy"]
)
```

```
model.fit(
    {
        "input_ids": encodings["input_ids"],
        "attention_mask": encodings["attention_mask"]
    },
    labels,
    epochs=3,
    batch_size=2,
    verbose=1
)
```

```
Epoch 1/3
1/1 [==============================] - 36s 36s/step - loss: 0.8075 - accuracy: 0.5000
Epoch 2/3
1/1 [==============================] - 2s 2s/step - loss: 0.5675 - accuracy: 0.5000
Epoch 3/3
1/1 [==============================] - 2s 2s/step - loss: 0.4422 - accuracy: 1.0000
<tf_keras.src.callbacks.History at 0x7deb54cf2de0>
```

```
test_text = ["I really enjoyed this!"]

test_enc = tokenizer(
    test_text,
    padding=True,
    truncation=True,
    max_length=32,
    return_tensors="tf"
)

outputs = model.predict({
    "input_ids": test_enc["input_ids"],
    "attention_mask": test_enc["attention_mask"]
})

pred = tf.argmax(outputs.logits, axis=1).numpy()[0]

print("Prediction:", "Positive" if pred == 1 else "Negative")
```

```
1/1 [==============================] - 8s 8s/step
Prediction: Positive
```